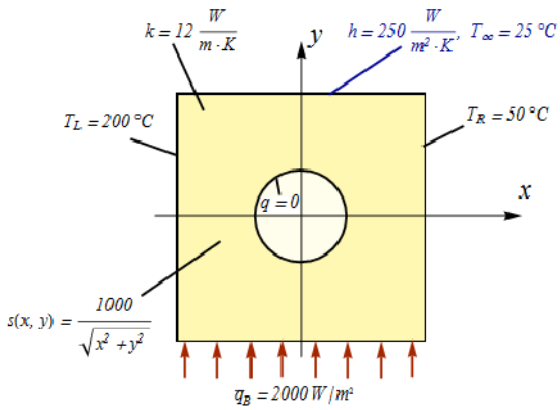
2) Matlab Cod e for a 2D Scalar Field FEA

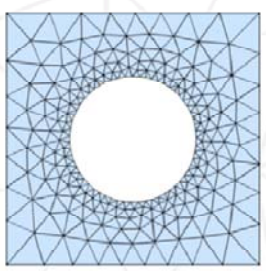
(a) Plate with a hole

Complete the partially-redacted 2D scalar field FEA code and solve the following 2D heat transfer problem for a plate with a hole. The plate thickness is 1, and the material is isotropic with a constant thermal conductivity (k). There is a distributed heat generation $s(x, y)$ acting throughout the plate. The surface of the hole is insulated, there is a prescribed temperature at the left and right faces, a prescribed heat flux into the plate at the bottom face, and a convection boundary condition at the top face.



The verification code will use a square domain, generated with the following mesh commands. Adjust the element size, as needed, to run well on your system.

```
% Example: (Rectangle with circular hole, refined at circle boundary)
    fd=@(p) ddiff(drectangle(p,-1,1,-1,1),dcircle(p,0,0,0.5));
    fh=@(p) 0.05+0.3*dcircle(p,0,0,0.5);
    [p,t]=distmesh2d(fd,fh,0.05,[-1,-1;1,1],[-1,-1;-1,1;1,-1;1,1]);
```



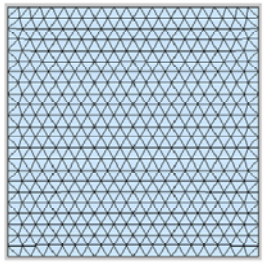
Submit:

- A contour plot of the temperature distribution in the plate showing heat flux vectors for two different mesh sizes.
- A plot of the nodal reactions at the left and right face, along with the total heat flux reaction at each face for both mesh sizes.
- The maximum and minimum temperature in the domain for each mesh size.
- A completed version of your code, with additional comments as needed.
- Briefly discuss solution quality. Is your code working correctly?

Verification

The verification code will use a square domain, generated with the following mesh commands. Adjust the element size, as needed, to run well on your system.

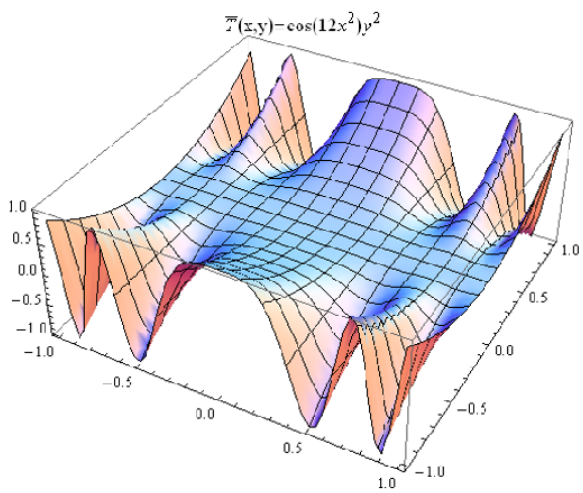
```
% Example: (Rectangle with uniform mesh)
fd=@(p) drectangle(p,-1,1,-1,1);
[p,t]=distmesh2d(fd,@huniform,0.1,[-1,-1;1,1],[-1,-1;-1,1;1,-1;1,1]);
```



Method of Manufactured Solutions

Complete the partially-redacted MMS code to solve for conduction in a square plate with generation. The body of your code (loop over elements) should be identical to the plate-with-a-hole code, except that you won't need to compute the Neumann or convection boundary terms. The boundary condition section will have to be modified to prescribe a spatially-varying Dirichlet boundary on all four faces.

Use a body source term $s(x, y)$ and spatially-varying Dirichlet boundary conditions so that the temperature distribution in the plate is $\overline{T}(x, y) = \cos(12\, x^2)\, y$

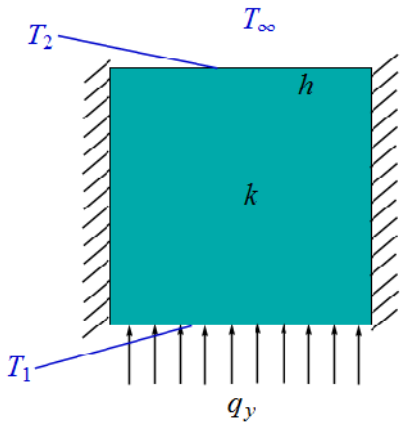


Submit:

- The expression for the heat source s(x,y)
- A surface plot of T_ex(x,y) and the compute s(x,y)
- A surface plot of the FEA solution for T_fe(x,y) and of the error (T_fe(x,y)-T_ex(x,y))
- An excerpt of your code showing how the Dirichlet boundary conditions are enforced.
- Briefly discuss solution quality. Is your code working correctly?

Convection Boundary

Modify your verification code to solve the following pseudo-1D convection verification problem. The boundary conditions on the top and bottom surface should be identical to the Plate-With-A-Hole simulation in part (a), but the left and right faces should be insulated. There is no generation term (i.e. $s(x, y) = 0$).



Submit:

- Analytical solution for the temperatures T_1 and T_2
- A contour plot of the temperature distribution in the plate showing heat flux vectors
- A comparison of the max and min nodal temperatures to the analytical solution.
- Briefly discuss solution quality. Is your code working correctly?